

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q17-0729

Cumberland Nj 138 kV  
July 15, 2018

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0729. The Interconnection Customer proposes a 86.2 MW battery energy storage facility interconnecting at the Cumberland Nj 138 kV substation in the New Jersey area. The request was received on January 17, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 282 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,408,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                          | Contingency                                         | Loading (%) | Rating (MVA) | DFAX (%) |
|---------------------------------------------|-----------------------------------------------------|-------------|--------------|----------|
| Springville to Conowingo 138kV              | Tombstone to Bixby Hill Substation 500kV line       | 218.9       | 360          | 5.4      |
| Sunbury 500 kV/345 kV TX                    | Coronet to Westwood Generation Llc 500kV            | 150.2       | 820          | 5.7      |
| Hastings Substation to Elk City Llc 345kV   | Little Chute to Fort Humbug 345kV                   | 130.3       | 1,005        | 5.4      |
| Burns to Coronado 345kV                     | Clear Springs 500/345kV Auto                        | 125.7       | 1,145        | 5.8      |
| W Gallatin to Walnut 345kV line             | Wallenpaupack 500-345 kV Transformer                | 118.9       | 1,477        | 6.8      |
| Hills Chapel to Burns 345kV                 | Clear Springs 500-345 kV TX                         | 116.2       | 1,468        | 6.5      |
| Wallenpaupack 500-345 kV Auto               | W Gallatin to Walnut 345kV                          | 115.1       | 1,606        | 6.6      |
| Irvine to Chiquita 345kV line               | Tombstone Substation to Marcus Hook Energy Lp 500kV | 113.4       | 1,462        | 16.7     |
| Connell to Chiquita 345kV                   | Tombstone to Marcus Hook Energy Lp 500kV            | 112.9       | 1,663        | 17.1     |
| Fayette to Ironwood Llc 345kV               | Brunner Island to Bixby Hill 500kV                  | 112.7       | 1,443        | 8.2      |
| Fort Humbug 500/345kV XFMR                  | Orchard Rd to FtHumbug Substation 500kV             | 108.4       | 1,409        | 6.3      |
| Owatonna to Moser Generating Sta 345kV line | Tombstone to Bixby Hill 500kV                       | 105.6       | 1,646        | 12.8     |
| Conell to Tombstone 345kV Line              | Tombstone to Marcus Hook Energy Lp 500kV            | 105.5       | 1,783        | 16.7     |

| Monitored Facility                                  | Contingency                                        | Loading (%) | Rating (MVA) | DFAX (%) |
|-----------------------------------------------------|----------------------------------------------------|-------------|--------------|----------|
| Strawberry Belt to Jh Warden 345kV L                | Coronet to Westwood Generation Llc 500kV Line      | 103.3       | 1,037        | 5.1      |
| N Allegheny Windpower Project to Wahiawa 345kV Line | Sunrise Ctr Substation to Merret 500kV line        | 103.2       | 1,447        | 5.5      |
| Ganso to HandsomeLake 500kV                         | Lions Rd to Piney Crk Project 500kV                | 100.2       | 2,387        | 10.3     |
| Medora to P.T.S 345kV                               | Sugar Hill to Allegheny Energy Units 3 4 & 5 345kV | 100.1       | 1,922        | 6.1      |

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                           | Required Upgrade                            | Cost Estimate | Allocation to Project |
|----------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Springville to Conowingo 138kV                     | Rebuild line with higher-capacity conductor | \$8,130,000   | 45% (\$3,657,000)     |
| Sunbury 500/345 kV Autotransformer                 | Replace transformer with higher-rated unit  | \$33,257,000  | 2% (\$646,000)        |
| Hastings to Elk City Llc 345kV                     | Rebuild line with higher-capacity conductor | \$17,589,000  | 1% (\$247,000)        |
| Burns to Coronado 345kV                            | Rebuild line with higher-capacity conductor | \$20,626,000  | 2% (\$389,000)        |
| West Gallatin to Walnut 345kV Line                 | Rebuild line with higher-capacity conductor | \$25,669,000  | 2% (\$473,000)        |
| Hills Chapel to Burns 345kV                        | Rebuild line with higher-capacity conductor | \$42,458,000  | 2% (\$813,000)        |
| Wallenpaupack 500-345 kV Auto                      | Replace transformer with higher-rated unit  | \$37,323,000  | 2% (\$695,000)        |
| Irvine to Chiquita Substation 345kV                | Rebuild line with higher-capacity conductor | \$50,705,000  | 3% (\$1,675,000)      |
| Connell to Chiquita 345kV                          | Rebuild line with higher-capacity conductor | \$44,939,000  | 3% (\$1,496,000)      |
| Fayette to Ironwood Llc 345kV                      | Rebuild line with higher-capacity conductor | \$34,641,000  | 1% (\$374,000)        |
| Fort Humbug 500/345kV XFMR                         | Replace transformer with higher-rated unit  | \$35,442,000  | 2% (\$544,000)        |
| Owatonna to Moser Generating Sta 345kV line        | Reconductor line                            | \$3,964,000   | 3% (\$112,000)        |
| Conell to Tombstone 345kV Line                     | Reconductor line                            | \$4,580,000   | 4% (\$165,000)        |
| Jh Warden to Strawberry Belt 345kV                 | Reconductor line                            | \$17,108,000  | 2% (\$375,000)        |
| North Allegheny Windpower Project to Wahiawa 345kV | Reconductor line                            | \$12,732,000  | 2% (\$193,000)        |
| Hansome Lake Substation to Ganso 500kV Line        | Reconductor line                            | \$48,925,000  | 1% (\$443,000)        |

| Facility              | Required Upgrade           | Cost Estimate | Allocation to Project |
|-----------------------|----------------------------|---------------|-----------------------|
| Medora to P.T.S 345kV | Reconductor line           | \$11,844,000  | 1% (\$111,000)        |
|                       | Total allocated to project |               | \$12,408,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.